From 472K Visits to 32K Purchases: A Conversion Funnel Case Study

Executive Summary

The analysis reveals that while the store generates healthy traffic, a significant "Yield Gap" exists in mobile conversion and cart abandonment. Although desktop users convert at a robust 8.5%, mobile users are falling off at more than double that rate. Furthermore, the Cart step is the single greatest point of failure, losing over 54% of potential buyers.

Key Findings:

- The Shopping Cart page has the highest bounce rate (54.83%), suggesting issues with hidden costs (shipping/tax) or a complex checkout flow.
- Bing Search (bsearch) consistently out-converts Google Search across both Brand and Non-Brand segments.
- Desktop users are nearly 3X more likely to convert than mobile users (8.50% vs. 3.09%).

Key recommendation:

Launch a mobile "Sticky Cart" and simplified checkout pilot immediately. Closing 25% of the mobile-desktop conversion gap would increase total orders by 12.5% with zero additional marketing spend.

Business Problem

The toy e-commerce retailer needs to analyze the full customer journey from first touchpoint to order confirmation in order to identify critical drop-off points, evaluate marketing channel performance across Google and Bing, and assess the impact of new product launches and device types on conversion rates.

Key Questions:

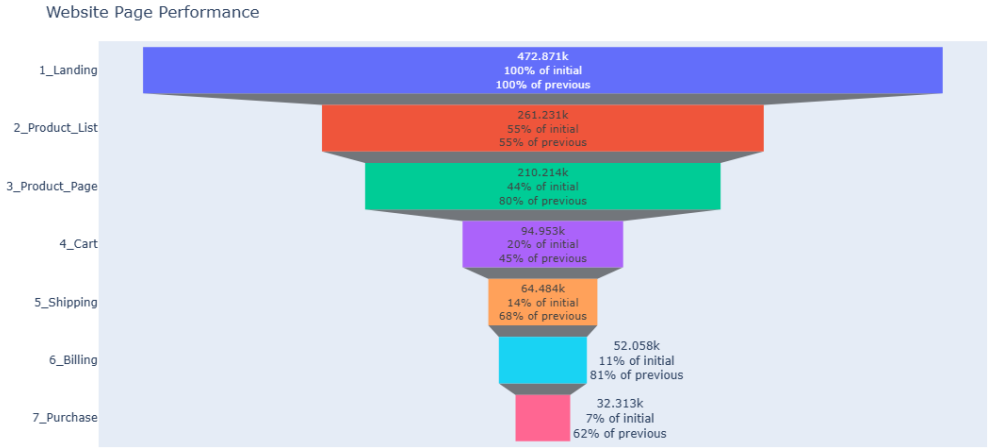
- Which pages have the highest bounce rates?
- Where do we lose the most customers?
- Does Google Search convert better than Bing Search? How does nonbrand vs brand performance differ?
- Is the mobile experience underperforming compared to desktop?
- Did the launch of the "Forever Love Bear" (Jan 2013) change the conversion rate at the /products step?

Data, Approach, & Methodology

- Data: Multi-year transaction and session logs (2012–2015) totaling nearly 500,000 sessions sourced from Maven Analytics
- Tools: Python (Pandas/NumPy) for data engineering and Jupyter for statistical modeling.
- Transformation:
    - Resolved 8,300 missing UTM parameters by identifying non-paid/organic paths.
    - Unified 5+ disparate landing page versions and 2 billing iterations into a single standardized 7-step funnel.
    - Created session-level binary flags for each funnel stage via pivot aggregation.

Key Findings

1. Overall, only 6.8% of Initial Sessions Convert to Purchase

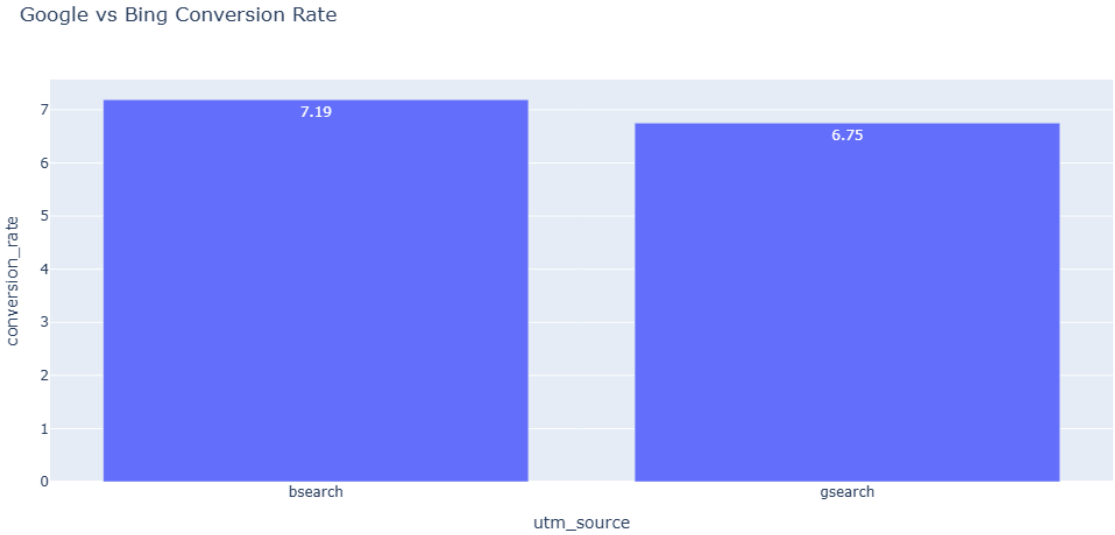


The vast majority of the 472,871 visitors leave before completing an order.

2. Shopping Cart Loses 54.8% of Users

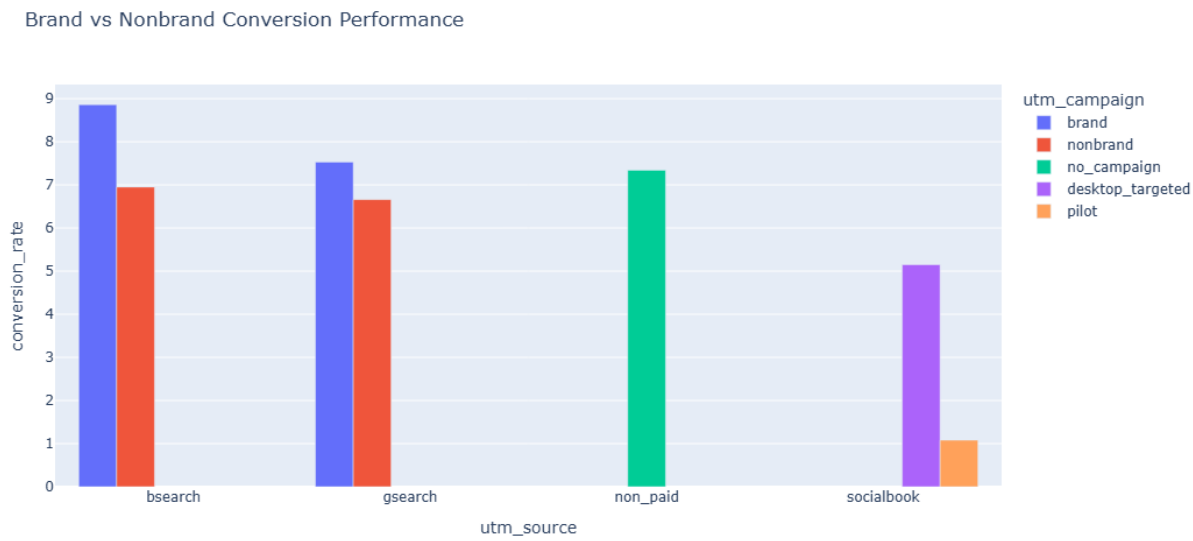
Only 45.2% of users who view a product proceed to cart. This single step loses 115,000 potential customers at purchase intent. The bounce rate at this stage (54.8%) exceeds billing (19.3%) and shipping (32.1%) combined.

3. Bing Outperforms Google on Conversion Rate Despite Lower Volume



Bing Search outperforms Google Search in conversion rate (7.19% vs. 6.75%), despite driving only 20% of the traffic volume. Bing delivers higher quality traffic even with significantly fewer sessions.

4. Brand Campaigns Dramatically Outperform Non-Brand Campaigns Across Both Search Engines



Bing brand converts at 8.86% vs. 6.95% non-brand; Google brand converts at 7.53% vs. 6.66% non-brand. Branded search intent is substantially more valuable.

5. Desktop Converts Nearly 3x Better than Mobile (8.50% Vs. 3.09%)

	device_type	total_sessions	total_orders	conversion_rate
	desktop	327027	27805	8.50
	mobile	145844	4508	3.09

Mobile sessions represent 31% of total traffic but only 14% of total orders, indicating a poor mobile user experience. The mobile experience is fundamentally broken. For every 100 mobile shoppers, 97 leave without buying.

6. "Forever Love Bear" Launch Lifted Product-to-Cart by +11.7%

	period	product_sessions	cart_sessions	product_to_cart_rate
0	Post-Launch	7019	2485	35.4
1	Pre-Launch	11593	3675	31.7

The "Forever Love Bear" launch improved product-to-cart conversion from 31.7% pre-launch to 35.4% post-launch, a 3.7 percentage point lift (+11.7% relative improvement). The new product positively influenced engagement at the product listing stage.

Strategic Recommendations

- **Reduce cart abandonment friction:** Address the Cart stage drop-off by simplifying checkout flow, surfacing shipping/tax costs earlier, and reducing unnecessary form steps.
- **Optimize mobile checkout experience: Prioritize** mobile-first improvements including one-click checkout, persistent cart functionality, and reduced input complexity across checkout pages.
- **Rebalance paid search allocation:** Increase investment in Bing Search and branded campaigns while reducing spend on lower-performing non-brand Google traffic.
- **Strengthen product-led funnel strategy:** Replicate successful product launch patterns (e.g., Forever Love Bear) by positioning high-performing products earlier in the funnel and testing product-led landing experiences.
- **Standardize funnel experience across page variants:** Consolidate legacy landing and billing page versions into a single optimized structure to reduce user confusion and fragmentation.
- **Enhance post-product page engagement:** Improve progression from product listing to cart through clearer value propositions, stronger CTAs, and reduced decision friction at product pages.